

HealthConnect - Week 7 Project Summary

Track: Data Science | Model Testing, Refinement & Validation

Goal

Week 7 focused on testing whether the Gradient Boosting model selected in Week 6 was stable on unseen patients. I also tested its weaknesses and refined the model where the evidence supported a change.

Testing and main result

Patient-grouped cross-validation gave a mean ROC-AUC of **0.67**, close to the Week 6 held-out result of **0.68**. This showed that the original result was reasonably stable. A training AUC of **0.77** versus test AUC of **0.68** also showed some overfitting.

Measure	Week 6	Week 7 tuned
Accuracy	65%	66%
No-show recall	65%	67%
No-show F1	65%	67%
ROC-AUC	0.68	0.72

Model refinement

I tuned the Gradient Boosting model to improve generalization. On the held-out test set, accuracy improved from **65% to 66%**, no-show recall from **65% to 67%**, F1 from **65% to 67%**, and ROC-AUC from **0.68 to 0.72**. The earlier train-test AUC gap was also reduced.

Segment and error analysis

Performance was not equal across all appointments. Recall was **78%** for patients with previous no-shows versus **58%** for those without. It was **94.3%** for bookings made more than 30 days ahead but only **23%** for bookings within 30 days. Specialist Consultations also had lower recall at about **53%**, compared with **80%** for Diagnostic Tests. Missed no-shows had an average booking lead time of about **17 days**, while correctly identified no-shows averaged about **44 days**.

Threshold test

I tested a lower threshold of **0.45**. Recall for appointments booked within 30 days improved from **23% to 37.7%**, but accuracy decreased and many short-lead no-shows were still missed. I therefore kept **0.50** as the main threshold and documented 0.45 as a higher-recall alternative.

Cross-track validation

A Data Analytics collaborator independently found a no-show rate of **36.84%** for appointments booked within 30 days and **60.31%** for bookings made more than 30 days ahead. This supported booking lead time as an important risk factor. It also showed that the model may rely too strongly on this signal when identifying no-shows.

Conclusion and Week 8

The tuned Gradient Boosting model is stronger than the Week 6 candidate and reached **0.72 ROC-AUC**. It is suitable as a moderate risk-screening and decision-support tool, but not as a standalone decision system. Short-lead no-shows and patients without previous no-show history remain harder to identify. Before Week 8, the final model should be prepared for wider HealthConnect integration and its remaining limitations should be clearly documented.